LoRa Pager – Portable Off-Grid Communication

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***Abstract* – This paper describes the selected parts, functionalities, and performance results of the LoRa pager prototype. LoRa pager is an off-grid peer-to-peer communicator that takes advantage of LoRa technology. The LoRa pager also uses the GNSS (global navigation satellite system) positioning to determine and send location.**

1 INTRODUCTION

In our modern world, we increasingly rely on many different types of networks (cellular networks, internet). These come with trivial advantages, for example, high bitrates, good coverage, seamless connection, and low power consumption on the user end. But it also comes with some disadvantages: networks are localized and fragile. Even a power outage can lead a city or a whole region to lose modern communication with the rest of the world. In the case of a natural catastrophe, for example, a flood, the most affected parts can lose all network-related connections. The only exceptions are radio amateurs who have expensive equipment and receive training. We wanted a more subtle, pocketable, and cheaper way of communicating long distances.

2 PARTS SELECTION

Searching the internet, we came across LoRa. LoRa is a somewhat new technology. It's a way of communication using spread spectrum modulations, a branch of CSS (Chirp spread spectrum) technology. LoRa uses an open ISM (Industrial, scientific, medical) frequency band. There is no license needed for transmitting on 433 MHz (ISM) in Europe (that being if the transmission is using less than 25 mW of power) [1]. Next, we needed a microcontroller. ESP32 fits all the needs and is widely available. For the user interface, we decided on a small OLED screen with three buttons and a piezo speaker to go with it.

3 FUNCTIONALITY

The pager functions in a way that it sends integers instead of strings of data over the air, and every integer has a pre-programmed text to go with it. We choose a page in the menu, and when we send it, the LoRa module only sends the integer that belongs to that page. Next, the receiving LoRa module receives that integer, and the text allocated to that integer gets displayed on the display of the receiving pager. The advantages of this type of communication are lower battery consumption since we need less time to send that two-byte integer than the dozen-byte string packet, ease of use, and the unnecessity of a functional text input method. But this design also poses a couple of drawbacks, one of which is that we only have a selected number of pages to send, thus limiting the information we can tell.

3.1 User Interface

With such a small form factor and the lack of buttons to navigate the user interface, we have programmed the pagers to have a selection menu. We can navigate different options for the pager's functionality in a menu. The menu has a layered setup, as seen in Figures 1 and 2.

Figure 1: Menu

Figure 2: Pages

3.2 GNSS

The pager has a GNSS receiver since geolocation is practical in dire situations (Figure 3). We can send or receive the location from one pager to another. The pagers can also act as an emergency beacon, sending the position with an emergency page every 10 seconds (Figure 4).

Figure 3: GNSS Info

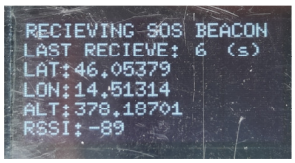

Figure 4: Receiving Emergency

4 HOUSING

The housing was 3D designed and 3D printed. It consists of 4 parts. The most significant is the main body, which has a hole for the USB micro connector. The bottom and the top part, the latter of which has four holes, three for buttons and the fourth one for the antenna. The fourth part is an acrylic glass that covers the front. The bottom and top were both screwed into place while the acrylic got glued (Figure 5).

Figure 5: Finished Pager

5 EXPECTATIONS

The world record for the LoRa transmission using 25 mW of power is 832 kilometers [2]. With that in mind, we were expecting a lot. Since 433 MHz is a high frequency and is very limited to the field of view, we expected our pagers would not work around big obstacles, such as hills or mountains.

6 RESULTS

We started testing the pagers with only the test code that pings one another. And at the beginning, the pagers were getting less than a hundred meters of range. The range increased after changing and calculating the best possible setting for LoRa. Testing the transmission range of the LoRa pagers if there is an obstacle in the way of the line of sight, has proven poorly. We had a transmitter that transmitted every 10 seconds about 200 meters away and 40 meters lower altitude from where we were standing on a small hill. The side facing the transmitter had reception, but we lost reception when we moved with our receiver just a few meters lower to the opposite side of the hill. After finishing the code for the pagers, we tested the range.

6.1 Short Distance

Short distance (~100m to ~2000m) line of sight pagers have no problem receiving. Signal strength (RSSI) ranges between -110 and -120 dBm. Trees, houses, and other small obstacles do not impact the signal strength.

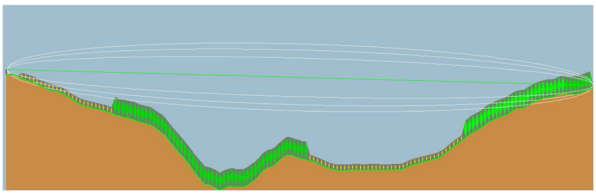

Figure 6: Side view of a 1509m successful transmission (RSSI -116)

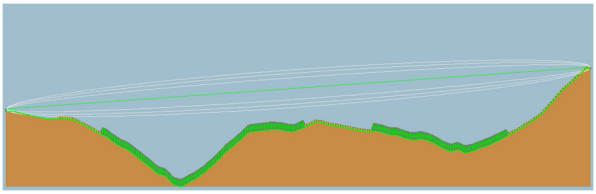

Figure 7: Side view of a 2412 m successful transmission (RSSI -114)

6.2 Long Distance

At longer distances (~2000m+) in line of sight, pagers have no problem receiving. RSSI is on the edge of pager capabilities, between -120 and -135 dBm. The estimated RSSI is -148 dBm [3], which is lower than what we managed to get, even when we barely received it. That RSSI would only be possible with the smallest bandwidth, but our pagers cannot handle anything below 62.5 kHz.

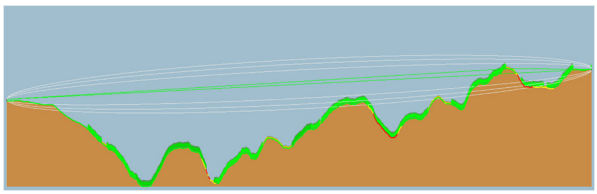

Figure 8: Side view of a not line of sight 7050 m successful transmission (RSSI -131)

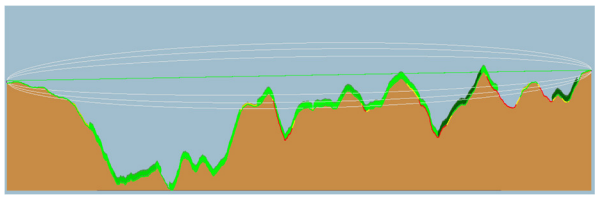

Figure 9: Side view of a 9231 m successful transmission (RSSI -132)

7 CONCLUSIONS

To sum up, we have built a portable, easy-to-use, and well-preforming off-grid communicator. The pagers have some advantages. For example, they can receive through small obstacles (houses, trees) and communicate far distances in line of sight. Pagers also have decent battery performance since they can last (on a full charge) a week in receiving mode and a day in a mode transmitting every 10 seconds and with receiving GNSS.

The pagers also have disadvantages, a critical one being the inability to communicate through larger obstacles. Because of this disadvantage, the usefulness in hilly areas is limited. There is room for improvement by using an STM32 microcontroller, optimizing the code, and integrating a mesh communication system.

References

[1] "LoRa" Wikipedia, https://en.wikipedia.org/wiki/LoRa, obtained: Aug 16th, 2023

[2] The Things Network https://www.thethingsnetwork.org/article/lorawan-world-record-broken-twice-in-single-experiment-1, obtained: Aug. 16th, 2023

[3] SX1276/77/78/79 – 137 MHz to 1020 MHz Low Power Long Range Transceiver, Semtech Corporation, Rev.5 – August 2016, Authorized Distributor: Mouser Electronics https://www.mouser.com/datasheet/2/761/sx1276-1278113.pdf, obtained Aug 16th, 2023